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Pass box

Abstract

A pass box configured to accommodate and decontaminate an article by concentrating a mist of decontamination agent on an article's surface and to reduce a duration of aeration operations. The box includes a decontamination chamber, an aeration chamber in which residue of the decontamination agent is removed from the article's surface, a decontamination agent supply device, a decontamination agent control device, a moving device configured to reposition the article, and an air supply and exhaust device. The decontamination agent supply device converts a chemical into the mist and supplies it into the decontamination chamber. The decontamination agent control device includes vibration boards that are adjacent to internal wall surfaces of the decontamination chamber and that are ultrasonically vibrated to generate sound flows in the vertical direction thereby pressing the mist with acoustic radiation to concentrate the mist on the external surfaces of the article.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application represents the US national stage entry from the International Application No. PCT/JP2020/011924, filed on Mar. 18, 2020, that claims priority from the Japanese Patent Application No. JP 2019-091088 filed on May 14, 2019. The disclosure of each of the above-identified patent documents is incorporated by reference herein.

TECHNICAL FIELD

(2) The present invention relates to a pass box associated with sterile environmental chambers such as an isolator and, more particularly, to a pass box capable of efficiently decontaminating an article conveyed to these sterile areas.

RELATED ART

(3) In manufacturing settings for pharmaceutical products and so forth, the indoor working area must inevitably be kept sterile. Particularly in cases where clean rooms (such as, for example, a working chamber for manufacturing pharmaceutical products) are decontaminated, advanced decontamination validation needs to be accomplished in accordance with Good Manufacturing Practice (GMP).

(4) The recent advanced technology in the field of regenerative medicine has led to wide use of incubators for cell culture. Such cell-based regenerative medicine techniques require steps for controlling, culturing and processing collected cells (Cell Processing), which are performed in plants referred to as “Cell Processing Center” (CPC). The CPC employs isolators as a working chamber that requires the highest cleanliness of grade A.

(5) A small chamber is employed as an isolator that allows an operator to work through gloves or a half-suit from the outside of the chamber. The isolator chamber is provided with an intake and exhaust device for maintaining a sterile state so as to receive no contaminants from the external environment. In addition, a sterile state is intended to be maintained when necessary equipment and articles are conveyed to the inside of an isolator in the sterile state from the external environment.

(6) For example, a small spare chamber for conveying, referred to as a “pass box”, is provided for an article to be conveyed to the inside of the isolator. An operator, who will convey an article to the inside of the isolator, first conveys the article to a pass box. In this case, a carry-in door between the isolator and the pass box is sealed. Subsequently, the carry-in door between the pass box and the external environment is sealed to decontaminate the article together with the inside of the pass box. After the pass box is completely decontaminated and a gas for decontamination or the like is removed, the carry-in door between the isolator and the pass box is opened to convey the article to the inside of the isolator.

(7) In recent years, hydrogen peroxide has widely been used (in the form of a gas or mist) to decontaminate a working chamber such as an isolator and a pass box (hereinafter referred to as a “room to be decontaminated”) and articles to be conveyed. Advantageously, hydrogen peroxide has a strong sterilization effect, and is inexpensively available and effectively utilized as an environmentally-friendly decontamination agent that is ultimately resolved into oxygen and water.

(8) In decontamination operations by hydrogen peroxide, a log spore reduction (LRD) value calculated by the logarithmic decrement of fungi can be 4 to 6 or more, indicating a sufficiently acceptable decontamination standard effect. Unfortunately, a decontamination work by hydrogen peroxide needs long duration of 45- to 120-minute operations including an aeration operation for removing hydrogen peroxide that is residual on an article or inside a pass box in the state of a

condensed film, or the like after decontamination, in addition to a decontamination operation.

(9) The following Patent Document 1 proposes a cell treatment device that employs no decontamination approaches by hydrogen peroxide in the field of regenerative medicine where articles are frequently conveyed to an isolator with the cleanliness maintained at grade A. The cell treatment device serially connects at two stages a pass box to an isolator of grade A. In the isolator and the pass box, the air pressure controls the air flow direction.

(10) First, at the first stage in the pass box to which an article has been conveyed from the external environment, an operator in the external environment decontaminates using gloves external surfaces of the article (a packaging bag that accommodates a sterilized container) by wiping out the same with an alcohol-containing cloth. Subsequently, the operator conveys the decontaminated packaging bag to the second stage in the pass box, and takes containers out of the packaging bag. Then, the containers are conveyed to the isolator of grade A.

CITATION LIST

Patent Literature

(11) Patent Document 1: WO-2017-069147 A

SUMMARY OF THE INVENTION

Technical Problem

(12) In fact, the invention in the above Patent Document 1 achieves shorter-time decontamination operations than conventional hydrogen peroxide-based techniques. Nevertheless, medical GMP standards fail to constantly assure an LRD value of 4 to 6 capable of achieving the grade A conveying environment. In addition, when the air pressure in the pass box is controlled, advanced control, such as air pressure cascade control, cannot readily be performed in such a small space. Therefore, shorter-time decontamination has been desired using conventionally widely used and highly reliable hydrogen peroxide.

(13) Thus, the present invention was made in view of the situation to solve the problems, and has an object to provide a pass box capable of accomplishing a decontamination effect with a proper amount of decontamination agent by concentrating a mist for decontamination on the surface of an article to be conveyed to grade A environment, and of achieving more efficient decontamination works by reducing the duration of aeration operations.

Solution to the Problem

(14) To solve the aforementioned problem, inventors of the present invention have carried out an extended investigation to find that a pass box is divided into a decontamination chamber and an aeration chamber, a hydrogen peroxide mist supplied to the pass box is miniaturized by employing ultrasonic vibration in a decontamination agent control device, and the hydrogen peroxide mist is allowed to concentrate on the surface of an article in the pass box. Based on that technique, the present invention was accomplished.

(15) Specifically, a pass box according to the present invention includes, according to recitation in claim 1, a pass box (100) used for conveying an article (P) to the inside of a working chamber or to a location outside the pass box with a sterile environment maintained, including a first chamber (30) for decontaminating external surfaces of the article with a decontamination agent, a second chamber (20) for removing the decontamination agent that is residual on the external surfaces of the decontaminated article, a decontamination agent supply device (70) for supplying a decontamination agent, a decontamination agent control device (60) for controlling the movement of the supplied decontamination agent, a moving device (50) for moving or repositioning the article, and an air supply and exhaust device for supplying and exhausting clean air, characterized in that the first chamber includes a first door (31) for conveying an article to the inside from the external environment, a supply port of the decontamination agent supply device, and an exhaust port (34) of the air supply and exhaust device, the second chamber includes a second door (25) provided on the boundary with the first chamber, a third door (21) for conveying the aerated article to the inside of the working chamber or to a location outside the pass box, and a supply port (23) of

the air supply and exhaust device, the decontamination agent supply device converts a chemical for decontamination into a mist for decontamination (M2), and supplies the same to the inside of the first chamber that accommodates the article through the supply port, the decontamination agent control device includes vibration boards (61, 62) disposed adjacent to internal wall surfaces of the first chamber, the vibration boards ultrasonically vibrate to generate sound flows (61b, 62b) from board surfaces (61a, 62a) by an ultrasound in the vertical direction, and the mist for decontamination supplied to the first chamber is pressed by acoustic radiation pressure to concentrate the mist for decontamination on the external surfaces of the article, the moving device moves/reposition the decontaminated article from the first chamber to the second chamber, and the air supply and exhaust device performs aeration for removing the decontamination agent that is residual on the external surfaces of the article by supplying clean air to the second chamber through the supply port, and discharges the air containing the aerated decontamination agent into the external environment through the exhaust port.

(16) Furthermore, the present invention includes, according to description in claim 2, the pass box according to claim 1, characterized in that the second chamber is located above the first chamber, the air supply and exhaust device is disposed such that a supply port thereof is open at an upper portion of the second chamber and an exhaust port thereof is open at a lower portion of the first chamber, characterized in that the clean air supplied from the air supply and exhaust device to the second chamber is converted into a unidirectional flow from the upper to lower portions of the first chamber through the external surfaces of the article from the upper portion of the second chamber, and discharged from the exhaust port that is open at the lower portion of the first chamber into the external environment while the external surfaces of the article and the inside of the first chamber are aerated when the article is in the second chamber for aeration.

(17) Moreover, the present invention includes, according to description in claim 3, the pass box according to claim 1 or 2, characterized in that the decontamination agent control device includes a plurality of vibration boards, characterized in that the plurality of vibration boards is arranged with the article in the first chamber placed therebetween and board surfaces thereof being opposite each other, and the resulting pressing force by the acoustic radiation pressure primarily concentrates in the direction of the article from each of the vibration boards, and the mist for decontamination is controlled to concentrate on the external surfaces of the article.

(18) Furthermore, the present invention includes, according to description in claim 4, the pass box according to claim 1 or 2, characterized in that the decontamination agent control device includes a plurality of vibration boards, characterized in that the plurality of vibration boards is arranged with the article in the first chamber placed therebetween and without board surfaces thereof being opposite each other, and the resulting pressing force by the acoustic radiation pressure primarily passes along a side surface of the article from each of the vibration boards, and the mist for decontamination is controlled to concentrate on the external surfaces of the article by moving so as to rotate inside the first chamber.

(19) Moreover, the embodiment of the present invention includes, according to description in claim 5, the pass box according to any one of claims 1 to 4, characterized in that the decontamination agent supply device includes a mist generation means, characterized in that the mist generation means is a spray nozzle such as a single-fluid nozzle or a two-fluid nozzle, or an ultrasonic mist generator such as an ultrasonic humidifier, a nebulizer or a piezo atomizer, and a primary mist of the decontamination agent generated by the mist generation means is supplied to the first chamber as a mist for decontamination through the supply port of the decontamination agent supply device included in the first chamber.

(20) Furthermore, the present invention is, according to description in claim 6, the pass box according to claim 5, characterized in that the decontamination agent supply device includes a mist microparticulating means (73), characterized in that the mist microparticulating means includes one or more vibration boards (73a, 73b), and the vibration boards ultrasonically vibrate to generate

sound flows (73e, 73f) from board surfaces (73c, 73d) by an ultrasound in the vertical, the primary mist (M1) of the decontamination agent generated by the mist generation means is subjected to ultrasonic vibration by the sound flows to further miniaturize the primary mist into a secondary mist (M2), and the generated secondary mist is supplied to the first chamber as a miniaturized mist for decontamination through the supply port of the decontamination agent supply device included in the first chamber.

(21) Moreover, the present invention includes, according to description in claim 7, the pass box according to any one of claims 1 to 6, characterized in that the second chamber includes one or more vibration boards (328) disposed adjacent to internal wall surfaces thereof, the vibration boards ultrasonically vibrate to generate sound flows (328b) from board surfaces (328a) by an ultrasound in the vertical direction, and the decontamination agent that is residual on the surface of the article is removed by allowing the clean air and the sound flow to act on the external surfaces of the decontaminated article to be aerated in the second chamber.

(22) Furthermore, the present invention includes, according to description in claim 8, the pass box according to any one of claims 1 to 7, characterized in that the vibration board includes a base (61c) and a plurality of transmitters (61e), characterized in that the plurality of transmitters is arranged on a planar surface of the base so as to be uniform in transmission directions, and the transmitters are operated in the same phase, whereby a sound flow is generated by a significantly directional ultrasound from the board surface of the vibration board in the vertical direction by mutually amplifying the ultrasounds of the plurality of transmitters in the front direction and mutually canceling out the ultrasounds of the plurality of transmitters in the lateral direction.

(23) Moreover, the embodiment of the present invention includes, according to description in claim 9, the pass box according to any one of claims 1 to 8, characterized in that the mist for decontamination supplied to the inside of the working chamber is further miniaturized by ultrasonic vibration generated from the vibration board.

(24) Furthermore, the embodiment of the present invention includes, according to description in claim 10, the pass box according to any one of claims 1 to 9, including a control device for changing the frequency and output of the ultrasound generated from the vibration board and/or for transmitting an ultrasound intermittently to control the position or moving speed of the mist for decontamination controlled to concentrate on the external surfaces of the article.

Advantageous Effects of the Invention

(25) According to the above configuration, the pass box of the present invention is used for conveying an article to the inside of a working chamber or to a location outside the pass box where the sterile environment is maintained, including a first chamber for decontaminating external surfaces of the article with a decontamination agent, a second chamber for removing the decontamination agent that is residual on the external surfaces of the decontaminated article, a decontamination agent supply device for supplying a decontamination agent, a decontamination agent control device for controlling the movement of the supplied decontamination agent, a moving device for moving the article, and an air supply and exhaust device for supplying and exhausting clean air.

(26) The first chamber includes a first door for conveying an article to the inside from the external environment, a supply port of the decontamination agent supply device, and an exhaust port of the air supply and exhaust device. The second chamber includes a second door provided on the boundary with the first chamber, a third door for conveying the aerated article to the inside of the working chamber or to a chosen location outside the pass box, and a supply port of the air supply and exhaust device. The decontamination agent supply device converts a chemical for decontamination into a mist for decontamination, and supplies the same to the inside of the first chamber that accommodates the article through the supply port. The decontamination agent control device includes vibration boards disposed adjacent to internal wall surfaces of the first chamber, the vibration boards ultrasonically vibrate to generate sound flows from board surfaces by an

ultrasound in the vertical direction, and the mist for decontamination supplied to the first chamber is pressed by acoustic radiation pressure to concentrate the mist for decontamination on the external surfaces of the article. The moving device repositions the decontaminated article from the first chamber to the second chamber. The air supply and exhaust device performs aeration for removing the decontamination agent that is residual on the external surfaces of the article by supplying clean air to the second chamber through the supply port, and discharges the air containing the aerated decontamination agent into the external environment through the exhaust port.

(27) Thus, according to the above configuration, the present invention can provide a pass box capable of accomplishing a decontamination effect with a proper amount of decontamination agent by concentrating a mist for decontamination on the surface of an article to be conveyed to grade A environment, and of achieving more efficient decontamination works by reducing the duration of aeration operations.

(28) According to the above configuration, the second chamber is located above the first chamber. The air supply and exhaust device is disposed such that a supply port thereof is open at an upper portion of the second chamber and an exhaust port thereof is open at a lower portion of the first chamber. Accordingly, the clean air supplied from the air supply and exhaust device to the second chamber is converted into a unidirectional flow from the upper to lower portions of the first chamber through the external surfaces of the article from the upper portion of the second chamber and the second door, and discharged from the exhaust port that is open at the lower portion of the first chamber into the external environment while the external surfaces of the article and the inside of the first chamber are aerated when the article is in the second chamber for aeration. Thus, the above operational advantage can more specifically be provided.

(29) According to the above configuration, the decontamination agent control device includes a plurality of vibration boards, these vibration boards are arranged with the article in the first chamber placed therebetween and board surfaces thereof being opposite each other, and the resulting pressing force by the acoustic radiation pressure primarily concentrates in the direction of the article from each of the vibration boards. Accordingly, the mist for decontamination is controlled to concentrate on the external surfaces of the article. Thus, the above operational advantage can more specifically be provided.

(30) According to the above configuration, the decontamination agent control device includes a plurality of vibration boards, these vibration boards are arranged with the article in the first chamber placed therebetween and without board surfaces thereof being opposite each other, and the resulting pressing force by the acoustic radiation pressure primarily passes along a side surface of the article from each of the vibration boards. Accordingly, the mist for decontamination is controlled to concentrate on the external surfaces of the article by moving so as to rotate inside the first chamber. Thus, the above operational advantage can more specifically be provided.

(31) According to the above configuration, the decontamination agent supply device includes a mist generation means. The mist generation means is a spray nozzle such as a single-fluid nozzle or a two-fluid nozzle, or an ultrasonic mist generator such as an ultrasonic humidifier, a nebulizer or a piezo atomizer. Accordingly, the decontamination agent supply device supplies a primary mist of the decontamination agent generated by the mist generation means to the first chamber as a mist for decontamination through the supply port of the decontamination agent supply device included in the first chamber. Thus, the above operational advantage can more specifically be provided.

(32) According to the above configuration, the decontamination agent supply device includes a mist microparticulating means, in addition to the mist generation means. The mist microparticulating means includes one or more vibration boards, and the vibration boards ultrasonically vibrate to generate sound flows from board surfaces by an ultrasound in the vertical direction. The primary mist of the decontamination agent generated by the mist generation means is subjected to ultrasonic vibration by the sound flows to further miniaturize or reduce in size the primary mist into a secondary mist that has smaller dimensions. Accordingly, the generated secondary mist is supplied

to the first chamber as a miniaturized mist for decontamination through the supply port of the decontamination agent supply device included in the first chamber. Thus, the above operational advantage can more specifically be provided.

(33) According to the above configuration, the second chamber includes one or more vibration boards disposed adjacent to internal wall surfaces thereof, the vibration boards ultrasonically vibrate to generate sound flows from board surfaces by an ultrasound in the vertical direction. Accordingly, the decontamination agent that is residual on the surface of the article is removed by allowing the clean air and the sound flow to act on the external surfaces of the decontaminated article to be aerated in the second chamber; in the second chamber, therefore, the decontaminated article is turned into an aerated article. Thus, the above operational advantage can more specifically be provided.

(34) According to the above configuration, the vibration board includes a base and a plurality of transmitters, and the plurality of transmitters is arranged on a plain surface of the base so as to be uniform in transmission directions, and the transmitters are operated in the same phase. Accordingly, a sound flow is generated by a significantly directional ultrasound from the board surface of the vibration board in the vertical direction by mutually amplifying the ultrasounds of the plurality of transmitters in the front direction and mutually canceling out the ultrasounds of the plurality of transmitters in the lateral direction. Thus, the above operational advantage can more specifically be provided.

(35) According to the above configuration, the mist for decontamination supplied to the inside of the working chamber is further miniaturized by ultrasonic vibration generated from the vibration board. Thus, the above operational advantage can more specifically be provided.

(36) According to the above configuration, the pass box may include a control device for changing the frequency and output of the ultrasound generated from the vibration board and/or for transmitting an ultrasound intermittently. Accordingly, the position or moving speed of the mist for decontamination controlled to concentrate on the external surfaces of the article is controlled. Thus, the above operational advantage can more specifically be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front view of a pass box according to a first embodiment;
- (2) FIG. 2 is a left side view of the pass box of FIG. 1;
- (3) FIG. 3 is a schematic front cross-sectional view showing the inside of (internal contents of) the pass box of FIG. 1 during decontamination;
- (4) FIG. 4 is a schematic plan cross-sectional view showing the inside of a first chamber of the pass box of FIG. 1 during decontamination;
- (5) FIG. 5 is a schematic front cross-sectional view showing a decontamination agent supply device included in the pass box of FIG. 1;
- (6) FIG. 6 is a schematic perspective view showing a plurality of ultrasonic speakers arranged in a speaker base in a vibration board included in the pass box of FIG. 1;
- (7) FIG. 7 is a schematic front cross-sectional view showing the inside of (internal contents of) the pass box of FIG. 1 during aeration;
- (8) FIG. 8 is a schematic plan cross-sectional view showing the inside of a first chamber of a pass box according to a second embodiment during decontamination; and
- (9) FIG. 9 is a schematic front cross-sectional view showing the inside of a pass box according to a third embodiment during aeration.

DETAILED DESCRIPTION

- (10) According to the idea of the present invention, a liquid decontamination agent or a

decontamination agent solution is allowed to act on an article to be decontaminated in the form of mist. Herein, “mist” is broadly interpreted as the state of a liquid droplet of a decontamination agent miniaturized and floating in the air, the state of a gas and a liquid agent of a decontamination agent in mixture, the state of the decontamination agent to repeat the change in phase between condensation and evaporation of a gas and a droplet, and the like. In terms of particle size as well, the mist is also broadly interpreted to include mists, fogs, and liquid droplets, which can be subclassified.

(11) Accordingly, the mist according to the present invention is categorized into a “mist” (the size may be defined as 10 μm or less) or a “fog” (the size may be defined as 5 μm or less), and a mist having a larger particle size. In the present invention, ultrasonic vibration converts even a mist, a fog and a liquid droplet sized 3 to 10 μm or more into equalized ultrafine particles 3 μm or less to provide high-level decontamination effects (later-described).

(12) The pass box according to the present invention will be described with reference to each of the embodiments. The present invention is not restricted to each of the following embodiments.

First Embodiment

(13) FIG. 1 is a front view of a pass box according to a first embodiment, and FIG. 2 is a left side view thereof. In FIGS. 1 and 2, a pass box **100** according to this first embodiment is configured by a pass box body **100a** placed on a floor G and an electrical/machine room **100b** joined to a wall portion of its right surface. In this first embodiment, a rear surface wall of the pass box **100** is linked to an isolator (not shown) maintained in grade A environment according to medical GMP standards.

(14) The pass box body **100a** is, in the order from an upper portion thereof, configured by an upper machine room **10**, a second chamber **20** (also referred to as “aeration chamber **20**”), a first chamber **30** (also referred to as “decontamination chamber **30**”) and a lower machine room **40**. The upper machine room **10** has therein a supply pipe (later-described) for supplying clean air to the aeration chamber **20**. The aeration chamber **20** is composed of a housing made of stainless metal plate, and a rear surface wall thereof is linked to a wall surface of an isolator (not shown) through an opening/closing door **21** (also referred to as “internal door **21**”) (see FIG. 2). The configuration of the inside of the aeration chamber **20** will be described later.

(15) The decontamination chamber **30** is composed of a housing made of stainless metal plate, and includes an opening/closing door **31** (also referred to as “external door **31**”) on a front wall thereof leading to the external environment and an opening/closing door (also referred to as “central door”; later-described) leading to the inside of the aeration chamber **20** (see FIG. 1). The details of the inside of the decontamination chamber **30** will be described later. The lower machine room **40** includes therein a decontamination agent supply device (later-described) for supplying a decontamination agent to the decontamination chamber **30** and an exhaust pipe (later-described) for discharging the air in the aeration chamber **20** and the decontamination chamber **30**.

(16) The electrical/machine room **100b** includes therein an air supply and exhaust device (not shown) for communicating with a supply pipe of the upper machine room **10** and an exhaust pipe of the lower machine room **40**, and a control device (not shown) and the like for controlling the pass box body **100a** such as a decontamination agent supply device or the like. The electrical/machine room **100b** also serves as a control board, and a front wall thereof is provided with a control monitor **S1** and a control panel **S2** (see FIG. 1).

(17) The linking state between the pass box body **100a** and the isolator and the structure of each opening/closing door (internal door **21**, external door **31**) are not particularly restricted, and the pass box is the same as conventional pass boxes in structure. The wall surface of the pass box linked to the isolator is not restricted to that on a rear surface wall, and the pass box may be linked to a right or left side wall or the like. The inside of the pass box may additionally be provided with an air supply and exhaust device for decontamination and aeration.

(18) Subsequently, using the pass box according to this first embodiment, operations for

decontaminating external surfaces of a package P that accommodates a plurality of sterilized containers and conveying the package to the inside of an isolator maintained in grade A environment will be described.

(19) FIG. 3 is a schematic front cross-sectional view showing the inside of the pass box according to this first embodiment during decontamination, and a cross-sectional view taken from line X-X of FIG. 2 (a partial cross-sectional view for the electrical/machine room **100b**). In FIG. 3, the upper machine room **10** is provided with a supply pipe **11** and an on-off valve **12** (each shown in a schematic diagram), each supplying clean air supplied from an air supply mechanism of an air supply and exhaust device (not shown) installed inside the electrical/machine room **100b** from a supply port **23** of an air supply and exhaust device that is open at the upper wall **22** of the aeration chamber **20** to the inside of the aeration chamber **20**. During the decontamination as shown in FIG. 3, the on-off valve **12** is controlled to be closed by a control device (not shown) of the pass box body **100a** to supply no clean air.

(20) The aeration chamber **20** is provided with the supply port **23** of the air supply and exhaust device (not shown) that is open at the above-described upper wall **22**, the internal door **21** mounted at an opening communicating the aeration chamber **20** and an isolator (not shown), the central door **25** provided at a communicating portion that is open at the boundary wall **24** between the aeration chamber **20** and the decontamination chamber **30**, and a filter fan unit **26** (FFU **26**) placed underneath the upper wall **22** of the aeration chamber **20**. Prior to the decontamination as shown in FIG. 3, the aeration chamber **20** is decontaminated beforehand and maintained in grade A or grade B environment.

(21) The central door **25** is controlled to be opened and closed by a control device (not shown) of the pass box body **100a**. Also, the FFU **26** is disposed such that clean air supplied from the supply port **23** of the air supply and exhaust device (not shown) during aeration is fed as a unidirectional flow from an upper portion of the aeration chamber **20** to a lower portion of the decontamination chamber **30**. During the decontamination as shown in FIG. 3, the internal door **21** and the central door **25** are air-tightly closed, and the FFU **26** is controlled to stop by a control device (not shown) of the pass box body **100a**.

(22) The decontamination chamber **30** is provided with an exhaust port **33** of an air supply and exhaust device (not shown) that is open at a lower wall **32**, a supply port **34** of a decontamination agent supply device **70** (later-described), a moving device **50** for moving a package P from the decontamination chamber **30** to the aeration chamber **20**, and a decontamination agent control device **60** (later-described in detail). In this first embodiment, the moving device **50** used is a lifting device **50**. The lifting device **50** is configured by a table **51** on which a package P is placed and an elevating/lowering unit (not shown) for allowing the table **51** to elevate and lower between the decontamination chamber **30** and the aeration chamber **20**, which is controlled to elevate and lower by a control device (not shown) of the pass box body **100a**. The shape of the table is not particularly restricted, and may be a mesh structure or the like for efficiently decontaminating a bottom surface portion of the package P. In addition, as a structure of providing the surface of a table with irregularities, the bottom surface portion of the package P may efficiently be decontaminated by ultrasonic vibration. Moreover, the package P may be provided in a suspended fashion without using a table. Meanwhile, the system and structure of an elevating/lowering unit may be any type such as manual hoisting, lift system or robot arm.

(23) The lower machine room **40** is provided with a decontamination agent supply device **70** for supplying a decontamination agent to the decontamination chamber **30**, which supplies the decontamination agent to the inside of the decontamination chamber **30** from the supply port **34** that is open at the lower wall **32** of the decontamination chamber **30**. During the decontamination as shown in FIG. 3, the decontamination agent is controlled to be supplied by a control device (not shown) of the pass box body **100a**. In addition, the lower machine room **40** is provided with an exhaust pipe **41**, an on-off valve **42**, and a catalyst unit **43** for resolving the decontamination agent

(each shown in a schematic diagram), and the air flowing out of the aeration chamber **20** and the decontamination chamber **30** is sucked out by suction from the exhaust mechanism of an air supply and exhaust device (not shown) installed inside the electrical/machine room **100b** to resolve the decontamination agent contained in the air and discharge the same into the external environment. During the decontamination as shown in FIG. **3**, the on-off valve **42** is controlled to be closed by a control device (not shown) of the pass box body **100a** to discharge no air.

(24) Subsequently, a decontamination operation as a first feature of the present invention will be described. The decontamination operation is performed in the decontamination chamber **30**. In FIG. **3**, the package **P** is conveyed from the external environment onto the table **51** of the moving device **50** at a center lower portion inside the decontamination chamber **30**. In this first embodiment, external surfaces of the package **P** are to be decontaminated. In FIG. **3**, inside the decontamination chamber **30**, the decontamination agent is supplied through the supply port **34** from the decontamination agent supply device **70** of the lower machine room **40**.

(25) Herein, the decontamination agent supply device **70** according to this first embodiment will be described. FIG. **5** is a schematic front cross-sectional view showing a decontamination agent supply device included in the pass box in FIG. **1** according to this first embodiment. In FIG. **5**, the decontamination agent supply device **70** includes a housing **71** made of stainless metal plate, a mist generation unit **72** for generating a primary mist, a mist microparticulating unit **73** for further miniaturizing (reducing in size) the generated primary mist into a secondary mist, and a supply port **74** for discharging the generated secondary mist.

(26) The mist generation unit **72** converts a decontamination agent into a primary mist and supplies the same to the mist microparticulating unit **73**. In this first embodiment, the mist generation unit **72** used is a two-fluid spray nozzle **72**, which is disposed on a bottom wall surface **71a** of the housing **71**. In this first embodiment, the decontamination agent used is a hydrogen peroxide solution (35 W/V %).

(27) The two-fluid spray nozzle **72** converts a hydrogen peroxide solution into a hydrogen peroxide solution mist **M1** as a primary mist by compressed air from a compressor (not shown) to supply the same to the mist microparticulating unit **73**. In the first embodiment, the mist generation unit used is a two-fluid spray nozzle, but it is not restricted thereto, and a mist generation unit and output are not particularly restricted.

(28) In this first embodiment, 2 vibration boards **73a**, **73b** are used as the mist microparticulating unit **73** and disposed against the inside of 2 side wall surfaces **71b**, **71c** opposite the housing **71** such that vibration surfaces **73c**, **73d** face in the horizontal direction toward an upper portion of the mist discharge port **72a** of the two-fluid spray nozzle **72**. The mist microparticulating unit **73** miniaturizes the hydrogen peroxide solution mist **M1** generated as the primary mist by the two-fluid spray nozzle **72** into a fine mist **M2** as a secondary mist. The details of miniaturizing will be described later.

(29) The supply port **74** is open on an upper wall surface **71d** of the housing **71**, and connected to the supply port **34** that is open at the lower wall **32** of the decontamination chamber **30**. Through the supply port **74** (the supply port **34** of the decontamination chamber **30**), the mist microparticulating unit **73** supplies the fine mist **M2** miniaturized as a secondary mist to the inside of the decontamination chamber **30**. The details of the vibration board **73** will be described later. In this first embodiment, a mist microparticulating unit is employed, but the unit is not restricted thereto in this invention, and the primary mist generated from the mist generation unit may be directly supplied to the inside of the decontamination chamber **30** without employing a mist microparticulating unit.

(30) Herein, a decontamination agent control device **60** will be described. FIG. **4** is a schematic plan cross-sectional view showing the inside of a first chamber (decontamination chamber) during decontamination, and a cross-sectional view taken from line Y-Y of FIG. **1**. In the front cross-sectional view in FIG. **3** and the plan cross-sectional view in FIG. **4** regarding the decontamination

chamber **30**, the decontamination agent control device **60** includes 2 vibration boards **61**, **62**. The vibration boards **61**, **62** are disposed inside the decontamination chamber **30** against left and right side walls **35**, **36** such that vibration surfaces **61a**, **62a** face horizontally inside the decontamination chamber **30**. These 2 vibration boards **61**, **62** are arranged by allowing board surfaces (vibration surfaces) thereof to be opposite each other (the board surfaces to face each other in front) (see FIG. **4**). The reason for arranging these vibration boards by allowing them to be opposite each other and the action of the fine mist M2 supplied to the inside of the decontamination chamber **30** will be described later.

(31) Herein, the vibration boards will be described. In this first embodiment, 2 vibration boards **61**, **62**, of the decontamination agent control device **60** and 2 vibration boards **73a**, **73b** of the mist microparticulating unit **73** are used. These vibration boards basically have the same structures and functions. Herein, the vibration board **61** of the decontamination agent control device **60** will be illustrated. FIG. **6** is a schematic perspective view showing a plurality of ultrasonic speakers arranged in a speaker base in a vibration board included in the pass box according to this first embodiment. In FIG. **6**, the vibration board **61** includes a base and a plurality of transmitters. In this first embodiment, the base used is a speaker base **61c**, and the transmitter used is an ultrasonic speaker **61e**. In this first embodiment, 25 ultrasonic speakers **61e** are arranged on a plain surface **61d** of the speaker base **61c** so as to be uniform in transmission direction of a vibration surface **61f** thereof (leftward as seen from the front shown). The number of ultrasonic speakers is not particularly restricted.

(32) In this first embodiment, the ultrasonic speaker **61e** used is an ultra-directional ultrasonic speaker. Specifically, ultrasonic speakers (DC 12V, 50 mA) of frequency modulation system transmitting a frequency near 40 KHz are used. The type, size, structure and output of the ultrasonic speaker are not particularly restricted. In the present invention, the vibration board included in the decontamination agent control device **60** or the mist microparticulating unit **73** is not restricted to an ultrasonic speaker, and the ultrasonic generation mechanism, frequency range and output are not particularly restricted.

(33) In this first embodiment, a plurality of (25) ultrasonic speakers **61e** are arranged so as to be uniform in transmission direction of the vibration surface **61f**, and the transmitters are operated in the same phase to mutually amplify ultrasounds from the plurality of ultrasonic speakers **61e** in the front direction and mutually cancel out ultrasounds from the plurality of ultrasonic speakers **61e** in the lateral direction. Consequently, the ultrasonic speakers **61e** arranged on the speaker base **61c** ultrasonically vibrate to generate a significantly directional sound flow traveling in the air from each of the vibration surfaces **61f** in the vertical direction. The frequency and output of each of the ultrasonic speakers **61e** of the decontamination agent control device **60** and the mist microparticulating unit **73** are controlled by a control device (not shown) of the pass box body **100a** to achieve efficient decontamination operations inside the decontamination chamber **30**.

(34) Subsequently, the action of the hydrogen peroxide solution mist in the decontamination agent supply device **70** and the decontamination agent control device **60** according to this first embodiment will be described. First, the action of the hydrogen peroxide solution mist in the decontamination agent supply device **70** will be described.

(35) In FIG. **5**, the hydrogen peroxide solution mist M1 generated as the primary mist by the two-fluid spray nozzle **72** is supplied to the mist microparticulating unit **73**. Ultrasonic vibration of the ultrasonic speaker **61e** of the 2 vibration boards **73a**, **73b** of the mist microparticulating unit **73** in this state generates significantly directional sound flows **73e**, **73f** traveling in the vertical direction from 2 vibration surfaces **73c**, **73d**, respectively. The sound flows **73e**, **73f** generated from the vibration surfaces **73c**, **73d** of the vibration boards **73a**, **73b** act on the hydrogen peroxide solution mist M1 discharged from the two-fluid spray nozzle **72**. The hydrogen peroxide solution mist M1 travels upward in the figure shown (in the direction of the supply port **74**) by a releasing pressure from the two-fluid spray nozzle **72**.

(36) At this time, the hydrogen peroxide solution mist M1 as a primary mist is miniaturized into a secondary mist M2 by ultrasonic vibration by the sound flows generated from the vibration surfaces 73c, 73d, and travels upward in the figure shown to be supplied to the inside of the decontamination chamber 30 communicating with an upper portion from the supply port 74 (connected to the supply port 34 that is open at the lower wall 32 of the decontamination chamber 30). In fact, since the fine mist M2 is miniaturized by ultrasonic vibration to have smaller particle sizes and larger surface areas, it is believed that the evaporation efficiency of mists is high, resulting in repeated evaporation and condensation to be supplied to the inside of the decontamination chamber 30 as a highly miniaturized mist.

(37) Subsequently, the action of the fine mist M2 supplied inside the decontamination chamber 30 will be described. In FIGS. 3 and 4, the vibration boards, 61, 62 placed at 2 side walls 35, 36 inside the decontamination chamber 30 allow vibration surfaces 61a, 62a thereof to face the central portion of the decontamination chamber 30 (in the same direction as the direction of the vibration surface 61f of the ultrasonic speaker 61e).

(38) Ultrasonic vibration of the ultrasonic speaker 61e of each vibration board in this state generates significantly directional sound flows 61b, 62b traveling in the vertical direction from the 2 vibration surfaces 61a, 62a, respectively. These sound flows 61b, 62b take in the fine mist M2 discharged from the mist microparticulating unit 73 through the supply port 34 to move the mist to the central portion of the decontamination chamber 30 each by pressing force from acoustic radiation pressure. In this case, the hydrogen peroxide solution mist M2 is further miniaturized into an ultrafine mist M3 by the ultrasonic vibration from the sound flows 61b, 62b.

(39) In FIG. 4, the 2 vibration boards 61, 62 are arranged with the package P interposed therebetween such that their vibration surfaces 61a, 62a are opposite. In this state, the package P conveyed to the central portion of the decontamination chamber 30 is interposed between the respective vibration surfaces 61a, 62a of the 2 vibration boards 61, 62. Inventors of the present invention found that in this state the ultrafine mist M3 further miniaturized by ultrasonic vibration from the sound flows 61b, 62b concentrates from both sides on external surfaces of the package P located at the central portion of the decontamination chamber 30 (see FIGS. 3 and 4). The reason has not clearly been identified, but this is probably because acoustic waves generated from each vibration board reach the article but only partially reflect off the same to primarily absorb or scatter on the surface of the article, thereby allowing the pressing force of the acoustic radiation pressure to concentrate in the article's direction.

(40) In fact, since the ultrafine mist M3 is ultra-miniaturized by ultrasonic vibration to have extremely smaller particle sizes and larger surface areas, it is believed that the evaporation efficiency of mists is high, resulting in repeated evaporation and condensation. The ultrafine mist M3 is a highly-miniaturized mist to concentrate on the external surfaces of the package P and form a uniform and thin condensed film on the external surfaces of the package P. Therefore, no excessive condensation occurs on internal wall surfaces of the decontamination chamber 30.

(41) Thus, the ultrafine mist M3 of hydrogen peroxide is subjected to constant ultrasonic vibration to concentrate on the periphery of the package P with repeated evaporation, condensation, and miniaturizing. Even on the external surfaces of the package P, the fine mist is subjected to constant ultrasonic vibration to cause repeated re-evaporation and condensation of a uniform and thin condensed film. Accordingly, it is believed that ultrafine particles of hydrogen peroxide 3 μ m or less and a hydrogen peroxide gas are subjected to phase change for coexistence on the periphery of the package P to provide high-level decontamination environment.

(42) Also, by repeated re-evaporation and condensation of the uniformly and thinly formed condensed film on the external surfaces of the package P, the concentration of a decontamination agent in a hydrogen peroxide solution mist can be increased and efficient decontamination can be performed with a small amount of decontamination agent. Such an efficient decontamination with a small amount of decontamination agent can significantly reduce the decontamination time.

Inventors of the present invention confirmed that the time required for decontaminating a package P, for specimens or the like, is approx. 3 to 5 minutes if sized about 10 to 100 mm, and approx. 10 to 20 minutes if sized about 300×300 mm. Furthermore, the secondary effect is that ultrasonic vibration and acoustic radiation pressure by the sound flows **61b**, **62b** can remove a deposit on external surfaces of the package P.

(43) Subsequently, an aeration operation as a second feature of the present invention will be described. The aeration operation is performed in the aeration chamber **20**. First, after the package P is completely decontaminated in the decontamination chamber **30**, the decontamination agent supply device **70** and the decontamination agent control device **60** are controlled to stop by a control device (not shown) of the pass box body **100a**. Subsequently, an air supply mechanism and an exhaust mechanism of an air supply and exhaust device (not shown) installed inside the electrical/machine room **100b** is operated by a control device. At this time, the central door **25** is controlled to be opened by the control device to communicate the aeration chamber **20** and the decontamination chamber **30**.

(44) FIG. 7 is a schematic front cross-sectional view showing the inside during aeration according to this first embodiment. In FIG. 7, the on-off valve **12** of the upper machine room **10** is opened to allow clean air from the air supply mechanism to be supplied to the inside of the aeration chamber **20** through the supply pipe **11** and the supply port **23**. In response thereto, the FFU **26** disposed underneath the supply port **23** is also operated. Also, the on-off valve **42** of the lower machine room **40** is opened to allow the air in the aeration chamber **20** and the decontamination chamber **30** to be discharged by suction from the exhaust mechanism to the external environment through the exhaust port **33** and the exhaust pipe **41**. At this time, the hydrogen peroxide contained in the discharged air is resolved by the catalyst unit **43**.

(45) In FIG. 7, the package P is moved to the aeration chamber **20** by an elevating/lowering unit (not shown) (the original position shown by a broken line in FIG. 7) while it is placed on the table **51** of the lifting device **50**. In this state, the clean air supplied through the supply port **23** of the aeration chamber **20** is converted into a unidirectional flow (as shown by arrows in FIG. 7) from an upper portion of the aeration chamber **20** to a lower portion of the decontamination chamber **30** by a fan of the FFU **26** and discharged into the external environment through the exhaust port **33** of the decontamination chamber **30** while the external surfaces of the package P and the inside of the decontamination chamber **30** are aerated. Accordingly, in this first embodiment, since aeration is performed by a unidirectional flow from the upper side to the lower side, the aeration chamber **20** is preferably disposed at an upper portion of the decontamination chamber **30**.

(46) As stated above, in the decontamination operation according to this first embodiment, the ultrafine mist M3 is a highly-miniaturized mist to concentrate on the external surfaces of the package P and form a uniform and thin condensed film on the external surfaces of the package P. Therefore, in the aeration operation in this first embodiment, the aeration chamber **20** is separated from the decontamination chamber **30**, and a condensed film to be removed is thin, resulting in improved aeration efficiency and reduced duration of decontamination. Inventors of the present invention confirmed that the effective time required for aerating a package P is approx. 1 to 2 minutes. Once the package P has been aerated, it is conveyed through the internal door **21** to the inside of an isolator with grade A maintained.

(47) Thus, according to this first embodiment, the present invention can provide a pass box capable of accomplishing a decontamination effect with a proper amount of decontamination agent by concentrating a mist for decontamination on the surface of an article to be conveyed to grade A environment, and of achieving more efficient decontamination works by reducing the duration of aeration operations.

Second Embodiment

(48) While the decontamination agent control device according to the above first embodiment relates to a pass box including in a decontamination chamber 2 vibration boards that are opposite

each other, this second embodiment relates to a pass box including in a decontamination chamber 2 vibration boards that are not opposite each other. FIG. 8 is a schematic plan cross-sectional view showing the inside of a first chamber (decontamination chamber) of the pass box according to a second embodiment during decontamination. A schematic front cross-sectional view showing the inside of the pass box according to this second embodiment during decontamination and a schematic front cross-sectional view showing the inside of the pass box during aeration are the same as in the above first embodiment (see FIGS. 3 and 7).

(49) In FIG. 8, a pass box **200** according to this second embodiment is configured by a pass box body **200a** and an electrical/machine room **200b** joined to a wall portion of its right surface. The pass box body **200a** is, as in the above first embodiment, in the order from an upper portion thereof, configured by an upper machine room, a second chamber (also referred to as “aeration chamber”), a first chamber **230** (also referred to as “decontamination chamber **230**”) and a lower machine room (the inside of the decontamination chamber **230** only shown in FIG. 8). The configurations of the upper machine room, the aeration chamber, the lower machine room and the electrical/machine room **200b** are the same as in the above first embodiment (see FIGS. 3 and 7).

(50) The decontamination chamber **230** is composed of a housing made of stainless metal plate, and includes an opening/closing door **231** (also referred to as “external door **231**”) on a front wall thereof leading to the external environment and an opening/closing door (not shown) leading to the inside of the aeration chamber **220**. In the plan cross-sectional view in FIG. 8, the decontamination agent control device **260** includes 2 vibration boards **261**, **262**. The vibration boards **261**, **262** are the same as the vibration boards **61**, **62** in the above first embodiment in structure (see FIG. 6). The frequency and output of these vibration boards **261**, **262** are also the same as those of the vibration boards in the above first embodiment.

(51) Subsequently, the action of a fine mist **M2** supplied to the inside of the decontamination chamber **230** will be described. Also in this second embodiment, the hydrogen peroxide solution mist is the fine mist **M2** supplied from the decontamination agent supply device as in the above first embodiment. In FIG. 8, the 2 vibration boards **261**, **262** are disposed inside the decontamination chamber **230** against side wall surfaces at 2 portions: a lower portion of a right wall surface **235** and an upper portion of a left wall surface **236** shown such that vibration surfaces **261a**, **262a** face horizontally inside the decontamination chamber **230**. These 2 vibration boards **261**, **262** are arranged without allowing board surfaces (vibration surfaces) thereof to be opposite each other.

(52) The vibration table **261** placed at the right lower portion shown inside the decontamination chamber **230** allows the vibration surface **261a** thereof to face in the left direction shown. Ultrasonic vibration of the ultrasonic speaker **61e** of each vibration board in this state allows a significantly directional sound flow **261b** traveling in the vertical direction (in the left direction shown) from the vibration surface **261a** to take in a fine mist **M2** discharged from the mist microparticulating unit through a supply port (not shown) to move the same in the direction of the sound flow **261b** (in the left direction shown) by pressing force from acoustic radiation pressure. In this case, the hydrogen peroxide solution mist **M2** is converted into an ultrafine mist **M3** that is further miniaturized by ultrasonic vibration from the sound flow **261b** and circulates so as to rotate inside the decontamination chamber **230**.

(53) Meanwhile, the vibration table **262** placed at the left upper portion shown inside the decontamination chamber **230** allows the vibration surface **262a** thereof to face in the right direction shown. Ultrasonic vibration of the ultrasonic speaker **61e** in this state allows a significantly directional sound flow **262b** traveling in the vertical direction (in the right direction shown) from the vibration surface **262a** to move the same in the direction of the sound flow **262b** (in the right direction shown) by pressing force from acoustic radiation pressure on the ultrafine mist **M3** miniaturized and sent by the sound flow **261b**. In this case, the hydrogen peroxide solution mist **M3** is converted into an ultrafine mist that is further stabilized by ultrasonic vibration from the

sound flow **262b** and circulates so as to rotate inside the decontamination chamber **230**.

(54) Thus, inside the decontamination chamber **230**, the ultrafine mist **M3** miniaturized and stabilized by the sound flows **261b** and **262b** circulate so as to rotate the mist in the curved arrow direction shown in FIG. **8** (clockwise). Inventors of the present invention found that the ultrafine mist **M3** miniaturized by ultrasonic vibration from the sound flows **261b** **262b** in this state circulates so as to rotate the mist by concentrating it around the external surfaces of the package **P** located at the central portion of the decontamination chamber **230** (see FIG. **8**). The reason has not clearly been identified, but this is probably because acoustic waves generated from each vibration board reach the article to diffract primarily along a side surface of the package **P** from each vibration board.

(55) In fact, since the ultrafine mist **M3** is ultra-miniaturized by ultrasonic vibration to have extremely smaller particle sizes and larger surface areas, it is believed that the evaporation efficiency of mists is high, resulting in repeated evaporation and condensation. The ultrafine mist **M3** is a highly-miniaturized mist to concentrate on the external surfaces of the package **P** and form a uniform and thin condensed film on the external surfaces of the package **P**. Therefore, no excessive condensation occurs on internal wall surfaces of the decontamination chamber **230**.

(56) Thus, the ultrafine mist **M3** of hydrogen peroxide is subjected to constant ultrasonic vibration to concentrate on the periphery of the package **P** with repeated evaporation, condensation, and miniaturizing. Even on the external surfaces of the package **P**, the fine mist is subjected to constant ultrasonic vibration to cause repeated re-evaporation and condensation of a uniform and thin condensed film. Accordingly, it is believed that ultrafine particles of hydrogen peroxide $3\text{ }\mu\text{m}$ or less and a hydrogen peroxide gas are subjected to phase change for coexistence on the periphery of the package **P** to provide high-level decontamination environment.

(57) Also, by repeated re-evaporation and condensation of the uniformly and thinly formed condensed film on the external surfaces of the package **P**, the concentration of a decontamination agent in a hydrogen peroxide solution mist can be increased and efficient decontamination can be performed with a small amount of decontamination agent. Such an efficient decontamination with a small amount of decontamination agent can significantly reduce the decontamination time.

Inventors of the present invention confirmed that the time required for decontaminating a package **P**, for specimens or the like, is approx. 3 to 5 minutes if sized about 10 to 100 mm, and approx. 10 to 20 minutes if sized about 300×300 mm. Furthermore, the secondary effect is that ultrasonic vibration and acoustic radiation pressure by the sound flows **261b**, **262b** can remove a deposit on external surfaces of the package **P**. An aeration operation in this second embodiment can be performed as in the above first embodiment.

(58) As stated above, in the decontamination operation according to this second embodiment, the ultrafine mist **M3** is a highly-miniaturized mist to concentrate on the external surfaces of the package **P** and form a uniform and thin condensed film on the external surfaces of the package **P**. Therefore, in the aeration operation according to this first embodiment, the aeration chamber is separated from the decontamination chamber, and a condensed film to be removed is thin, resulting in improved aeration efficiency and reduced duration of decontamination. Inventors of the present invention confirmed that the effective time required for aerating a package **P** is approx. 1 to 5 minutes.

(59) Thus, according to this second embodiment, the present invention can provide a pass box capable of accomplishing a decontamination effect with a proper amount of decontamination agent by concentrating a mist for decontamination on the surface of an article to be conveyed to grade A environment, and of achieving more efficient decontamination works by reducing the duration of aeration operations.

Third Embodiment

(60) In the above first embodiment, no vibration board is used during aeration, but in this third embodiment, the case where a vibration board is used even during aeration will be described. FIG.

9 is a schematic front cross-sectional view showing the inside of a pass box according to this third embodiment during aeration. An operation for decontaminating the pass box according to the third embodiment can be performed as in the above first and second embodiments except the way a vibration board provided in an aeration chamber is operated.

(61) In FIG. **9**, a pass box **300** according to this third embodiment is configured by a pass box body **300a** and an electrical/machine room **300b** joined to a wall portion of its right surface. The pass box body **300a** is, as in the above first embodiment, in the order from an upper portion thereof, configured by an upper machine room **310**, a second chamber **320** (also referred to as “aeration chamber **320**”), a first chamber **330** (also referred to as “decontamination chamber **330**”), and a lower machine room **340**. The configurations of the upper machine room **310**, the decontamination chamber **330**, the lower machine room **340** and the electrical/machine room **300b** are the same as in the above first embodiment (see FIGS. **3**, **4**, **5** and **7**).

(62) In FIG. **9**, the upper machine room **310** is, as in the above first embodiment, provided with a supply pipe **311** and an on-off valve **312** (each shown in a schematic diagram), each supplying the clean air supplied from an air supply mechanism of an air supply and exhaust device (not shown) installed inside the electrical/machine room **300b** from the supply port **323** of an air supply and exhaust device that is open at the upper wall **322** of the aeration chamber **320** to the inside of the aeration chamber **320**. During the aeration as shown in FIG. **9**, the on-off valve **312** is controlled to be open by a control device (not shown) of the pass box body **300a** to supply clean air.

(63) The aeration chamber **320** is provided with the supply port **323** of the air supply and exhaust device (not shown) that is open at the above-described upper wall **322**, the internal door **321** mounted at an opening communicating the aeration chamber **320** and an isolator (not shown), a central door **325** provided at a communicating portion that is open at the boundary wall **324** between the aeration chamber **320** and the decontamination chamber **330**, and a filter fan unit **326** (FFU **326**) placed underneath the upper wall **322** of the aeration chamber **320**. In this third embodiment, one vibration board **328** is disposed inside the aeration chamber **320** against a left side wall **327** such that a vibration surface **328a** faces horizontally inside the aeration chamber **320**.

(64) The aeration operation in this third embodiment is performed in the aeration chamber **320**. First, after a package **P** is completely decontaminated in the decontamination chamber **330**, a decontamination agent supply device **370** and a decontamination agent control device **360** are controlled to stop by a control device (not shown) of the pass box body **300a**. Subsequently, an air supply mechanism and an exhaust mechanism of an air supply and exhaust device (not shown) installed inside the electrical/machine room **300b** are operated by a control device. At this time, the central door **325** is opened by the control device to communicate the aeration chamber **320** and the decontamination chamber **330**.

(65) In FIG. **9**, the on-off valve **312** of the upper machine room **310** is opened to allow the clean air from the air supply mechanism to be supplied to the inside of the aeration chamber **320** through the supply pipe **311** and the supply port **323**. In response thereto, the FFU **326** disposed underneath the supply port **323** is also operated. In addition, the on-off valve **342** of the lower machine room **340** is opened to allow the air in the aeration chamber **320** and the decontamination chamber **330** to be discharged by suction from the exhaust mechanism to the external environment through the exhaust port **333** and the exhaust pipe **341**. At this time, the hydrogen peroxide contained in the discharged air is resolved by the catalyst unit **343**. In this third embodiment, the vibration board **328** is operated during aeration.

(66) In FIG. **9**, the package **P** is moved to the aeration chamber **320** by an elevating/lowering unit (not shown) (the original position shown by a broken line in FIG. **9**) while it is placed on the table **351** of the lifting device **350**. In this state, the clean air supplied through the supply port **323** of the aeration chamber **320** is converted into a unidirectional flow (as shown by arrows in FIG. **9**) from an upper portion of the aeration chamber **320** to a lower portion of the decontamination chamber **330** by a fan of the FFU **326** and discharged into the external environment through the exhaust port

333 of the decontamination chamber **330** while the external surfaces of the package P and the inside of the decontamination chamber **330** are aerated. Accordingly, in this third embodiment, since aeration is performed by a unidirectional flow from the upper side to the lower side, the aeration chamber **320** is preferably disposed at an upper portion of the decontamination chamber **330**.

(67) In an aeration operation according to this third embodiment, an ultrafine mist M3 forms a uniform and thin condensed film on the external surfaces of the package P. The supply of the unidirectional flow of clean air by the action of the FFU **326** fan in this state and ultrasonic vibration of the ultrasonic board **328** generate significantly directional sound flows **328b** traveling in the vertical direction from the vibration surface **328a**. The sound flow **328b** allows pressing forces by ultrasonic vibration and acoustic radiation pressure to act on the external surfaces of the package P. Accordingly, a thin condensed film formed on the external surfaces of the package P is vibrated, and the condensed film is proactively dried by the action of clean air, resulting in higher aeration effects. Once the package P has been aerated, it is conveyed through the internal door **321** to the inside of an isolator with grade A maintained.

(68) Therefore, in the aeration operation according to this third embodiment, the aeration chamber **320** is separated from the decontamination chamber **330**, and the action of the vibration board further improves the aeration efficiency and reduce the duration of aeration.

(69) Thus, according to this third embodiment, the present invention can provide a pass box capable of accomplishing a decontamination effect with a proper amount of decontamination agent by concentrating a mist for decontamination on the surface of an article to be conveyed to grade A environment, and of achieving more efficient decontamination works by reducing the duration of aeration operations.

(70) The present invention is achieved by not only each of the above embodiments, but also by the following various alternatives.

(71) (1) In each of the above embodiments, a decontamination agent supply device used is a combination of a mist generation unit and a mist microparticulating unit. However, the decontamination agent supply device is not restricted thereto, and a decontamination agent mist may directly be supplied from a mist generation unit to a decontamination chamber.

(72) (2) In each of the above embodiments, a mist generation unit of a decontamination agent supply device used is a two-fluid spray nozzle. However, the mist generation unit is not restricted thereto, and it may be other spray nozzle such as a single-fluid nozzle, or an ultrasonic mist generator such as an ultrasonic humidifier, a nebulizer or a piezo atomizer. Also, a combination of a plurality of mist supply devices may be used.

(73) (3) In each of the above embodiments, vibration boards used in a mist microparticulating unit, a mist control device and an aeration chamber include a plurality of ultrasonic speakers arranged in a speaker base. However, the vibration board is not restricted thereto, and any type of vibration board may be used so long as it includes a Langevin type transducer fixed to a stainless steel having a constant area or a board surface for ultrasonic vibration.

(74) (4) In each of the above embodiments, vibration boards used in a mist microparticulating unit, a mist control device and an aeration chamber include a plurality of ultrasonic speakers arranged in a speaker base so as to be uniform in transmission directions of the ultrasonic speakers, and the ultrasonic speakers are operated in the same phase. However, the vibration boards are not restricted thereto, and a plurality of ultrasonic speakers may be operated in a different phase.

(75) (5) In each of the above embodiments, a decontamination agent used is a hydrogen peroxide solution (H.sub.2O.sub.2 solution). However, the decontamination agent is not restricted thereto, and it may be any type of decontamination agent so long as it is liquid.

(76) (6) In each of the above embodiments, 2 vibration boards are arranged on side walls of the decontamination chamber. However, the configuration is not restricted thereto, and vibration boards may be arranged on 1 to 6 vibration boards out of 6 boards including other side walls, a top

wall surface and a bottom wall surface.

(77) (7) In the above first embodiment, 2 vibration boards are arranged so as to be opposite each other in the decontamination chamber. However, the positional relationship is not restricted thereto, and ultrasonic reflection boards may be arranged so as to be opposite each other relative to one vibration board.

(78) (8) In the above second embodiment, the circulating direction of a decontamination agent mist is horizontal direction in cases where 2 vibration boards are arranged on 2 side walls so as not to be opposite each other in the decontamination chamber. However, the circulation direction is not restricted thereto, and the circulating direction of a decontamination agent mist may be vertical direction in cases where 2 vibration boards may be arranged on a top wall and a bottom wall so as not to be opposite each other.

(79) (9) In the above second embodiment, the circulating direction of a decontamination agent mist is horizontal direction in cases where 2 vibration boards are arranged on 2 side walls so as not to be opposite each other. However, the circulation direction is not restricted thereto, and the circulating direction of a decontamination agent mist may be horizontal direction in cases where 4 vibration boards may be arranged on 4 side walls so as not to be opposite each other.

(80) (10) In each of the above embodiments, a moving device for moving a package from a decontamination chamber to an aeration chamber employed is a lifting device. However, the moving device is not restricted thereto, and other moving devices such as a lift and a robot arm may be used.

(81) (11) In each of the above embodiments, a decontamination chamber and an aeration chamber are disposed in the vertical direction. However, the arrangement is not restricted thereto, and they may be arranged laterally so long as the flow of the air in the both chambers during aeration can be controlled.

REFERENCE SIGNS LIST

(82) **100, 200, 300** . . . Pass box, **100a, 200a, 300a** . . . Pass box body, **100b, 200b, 300b** . . . Electrical/machine room, **10, 310** . . . Upper machine room, **11, 311** . . . Supply pipe, **12, 312** . . . On-off valve, **20, 320** . . . Second chamber (Aeration chamber), **21, 321** . . . Opening/closing door (Internal door), **22, 322** . . . Upper wall, **23, 323** . . . Supply port, **24, 324** . . . Boundary wall, **25, 325** . . . Opening/closing door (Central door), **26, 326** . . . Filter fan unit (FFU), **327** . . . Side wall, **328** . . . Vibration board, **328a** . . . Vibration surface, **328b** . . . Sound flow, **30, 230, 330** . . . First chamber (Decontamination chamber), **31, 231** . . . Opening/closing door (External door), **32** . . . Lower wall, **33, 333** . . . Exhaust port, **34, 334** . . . Supply port, **35, 36** . . . Side wall, **40, 340** . . . Lower machine room, **41, 341** . . . Exhaust pipe, **42, 342** . . . On-off valve, **43, 343** . . . Catalyst unit, **50, 350** . . . Moving device, **51, 351** . . . Table, **60** . . . Decontamination agent control device, **61, 62, 261, 262** . . . Vibration board, **61a, 62a, 261a, 262a** . . . Vibration surface, **61b, 62b, 261b, 262b** . . . Sound flow, **61c** . . . Speaker base, **61d** . . . plane or planar surface of speaker base, **61e** . . . Ultrasonic speaker, **61f** . . . Vibration surface, **70, 370** . . . Decontamination agent supply device, **71** . . . Housing, **71a** . . . Bottom wall surface, **71b, 71c** . . . Side wall surface, **71d** . . . Upper wall surface, **72** . . . Mist generation unit (Two-fluid spray nozzle), **72a** . . . Discharge port, **73** . . . Mist microparticulating unit, **73a, 73b** . . . Vibration board, **73c, 73d** . . . Vibration surface, **73e, 73f** . . . Sound flow, **74** . . . Supply port, **M1** . . . Hydrogen peroxide solution mist, **M2** . . . Fine mist, **M3** . . . Ultrafine mist, **S1** . . . Control monitor, **S2** . . . Control panel, **G** . . . Floor.

Claims

1. A pass box configured to convey an article therethrough and deliver said article to a location outside the pass box, the pass box comprising: a first chamber dimensioned to receive the article therein and configured to decontaminate external surfaces of the article with a decontamination agent, a second chamber wholly located above the first chamber and dimensioned to receive the

article therein and configured to remove the decontamination agent that is residual on the external surfaces of the decontaminated article, a decontamination agent supply device configured to supply the decontamination agent, a decontamination agent control device configured to control a movement of the supplied decontamination agent, a moving device configured to move the article placed thereon, and an air supply and exhaust device configured to supply and exhaust clean air, wherein: the first chamber includes a first door dimensioned to convey the article therethrough to an inside of the first chamber from an external environment, a first supply port of the decontamination agent supply device, and an exhaust port of the air supply and exhaust device, the second chamber includes a second door provided on a boundary with the first chamber and dimensioned to receive the article that has been decontaminated in the first chamber, a third door dimensioned to convey the article that has been aerated in the second chamber to the location outside the pass box, and a second supply port of the air supply and exhaust device, the decontamination agent supply device is configured to convert the decontamination agent into a mist for decontamination, and to supply said mist through the first supply port inside the first chamber, the decontamination agent control device includes vibration boards disposed adjacent to internal wall surfaces of the first chamber, wherein the vibration boards are configured to ultrasonically vibrate to generate sound flows from board surfaces in a vertical direction to press the mist supplied to the first chamber with acoustic radiation pressure thereby concentrating the mist on the external surfaces of the article, the moving device is configured to move the decontaminated article from the first chamber to the second chamber, which is wholly located above the first chamber, through the second door, and the air supply and exhaust device is configured to perform aeration to remove a residue of the decontamination agent from the external surfaces of the article by supplying clean air to the second chamber through the second supply port, and to discharge the air containing an aerated decontamination agent into the external environment through the exhaust port.

2. The pass box according to claim 1, wherein the air supply and exhaust device is disposed such that the second supply port thereof is open at an upper portion of the second chamber and the exhaust port thereof is open at a lower portion of the first chamber, wherein the pass box is configured to convert the clean air supplied from the air supply and exhaust device to the second chamber into a unidirectional flow from the upper to lower portions of the first chamber through the external surfaces of the article from the upper portion of the second chamber, and to discharge said clean air from the exhaust port that is open at the lower portion of the first chamber into the external environment while the external surfaces of the article and the inside of the first chamber are aerated when the article is in the second chamber.

3. The pass box according to claim 2, wherein the plurality of the vibration boards is arranged to have the article in the first chamber placed therebetween and board surfaces thereof being opposite each other to primarily concentrate the resulting pressing force with the acoustic radiation pressure in a direction of the article from each of the vibration boards.

4. The pass box according to claim 3, wherein: the decontamination agent supply device includes a mist generation means, wherein the mist generation means is (i) a spray nozzle including a single-fluid nozzle or a two-fluid nozzle, or (ii) an ultrasonic mist generator including an ultrasonic humidifier, a nebulizer, or a piezo atomizer, and a primary mist of the decontamination agent generated by the mist generation means is supplied to the first chamber as the mist for decontamination through the first supply port of the decontamination agent supply device included in the first chamber.

5. The pass box according to claim 4, wherein: the decontamination agent supply device includes a mist microparticulating means, wherein the mist microparticulating means includes one or more vibration boards configured to ultrasonically vibrate to generate sound flows from board surfaces by an ultrasound in the vertical direction, the pass box configured to subject the primary mist of the decontamination agent generated by the mist generation means to ultrasonic vibration by the sound

flows to further miniaturize the primary mist into a secondary mist, and to supply the secondary mist to the first chamber as a miniaturized mist for decontamination through the first supply port of the decontamination agent supply device included in the first chamber.

6. The pass box according to claim 5, wherein: the second chamber includes one or more vibration boards disposed adjacent to internal wall surfaces thereof, the vibration boards are configured to ultrasonically vibrate to generate sound flows from board surfaces by an ultrasound in the vertical direction, and to remove the decontamination agent that is residual on the surfaces of the article by passing the clean air and the sound flow in the second chamber to act on the external surfaces of the decontaminated article to be aerated.

7. The pass box according to claim 6, wherein: the vibration board includes a base and a plurality of transmitters, wherein transmitters from the plurality of transmitters are arranged on the planar surface of the based spatially-uniformly in transmission directions, and the transmitters are operated in the same phase, whereby a sound flow is generated by a significantly directional ultrasound from the board surface of the vibration board in the vertical direction by mutually amplifying the ultrasounds of the plurality of transmitters in the front direction and mutually canceling out the ultrasounds of the transmitters in a lateral direction.

8. The pass box according to claim 7, configured to further miniaturize the mist for decontamination supplied to the inside of the first chamber with ultrasonic vibration generated from the vibration board.

9. The pass box according to claim 8, comprising a control device configured to change the frequency and output of the ultrasound generated from the vibration board and/or to transmit the ultrasound intermittently to control a position or a moving speed of the mist for decontamination to concentrate said mist on the external surfaces of the article.

10. The pass box according to claim 2, wherein: the decontamination agent control device includes a plurality of the vibration boards, wherein the plurality of the vibration boards is arranged such that the article in the first chamber is placed therebetween and without board surfaces thereof being opposite each other, and to direct the resulting pressing force with the acoustic radiation pressure to primarily pass along a side surface of the article from each of the vibration boards, and to control the mist for decontamination to concentrate the mist on the external surfaces of the article by moving the mist so as to rotate the mist inside the first chamber.

11. The pass box according to claim 10, wherein: the decontamination agent supply device includes a mist generation means, wherein the mist generation means is (i) a spray nozzle including a single-fluid nozzle or a two-fluid nozzle, or (ii) an ultrasonic mist generator including an ultrasonic humidifier, a nebulizer, or a piezo atomizer, and a primary mist of the decontamination agent generated by the mist generation means is supplied to the first chamber as the mist for decontamination through the first supply port of the decontamination agent supply device included in the first chamber.

12. The pass box according to claim 11, wherein: the decontamination agent supply device includes a mist microparticulating means, wherein the mist microparticulating means includes one or more vibration boards configured to ultrasonically vibrate to generate sound flows from board surfaces by an ultrasound in the vertical direction, the pass box configured to subject the primary mist of the decontamination agent generated by the mist generation means to ultrasonic vibration by the sound flows to further miniaturize the primary mist into a secondary mist, and to supply the secondary mist to the first chamber as a miniaturized mist for decontamination through the first supply port of the decontamination agent supply device included in the first chamber.

13. The pass box according to claim 12, wherein: the second chamber includes one or more vibration boards disposed adjacent to internal wall surfaces thereof, the vibration boards are configured to ultrasonically vibrate to generate sound flows from board surfaces by an ultrasound in the vertical direction, and to remove the decontamination agent that is residual on the surfaces of the article by passing the clean air and the sound flow in the second chamber to act on the external

surfaces of the decontaminated article to be aerated.

14. The pass box according to claim 13, wherein: the vibration board includes a base and a plurality of transmitters, wherein transmitters from the plurality of transmitters are arranged on the planar surface of the based spatially-uniformly in transmission directions, and the transmitters are operated in the same phase, whereby a sound flow is generated by a significantly directional ultrasound from the board surface of the vibration board in the vertical direction by mutually amplifying the ultrasounds of the plurality of transmitters in the front direction and mutually canceling out the ultrasounds of the transmitters in a lateral direction.

15. The pass box according to claim 14, configured to further miniaturize the mist for decontamination supplied to the inside of the first chamber ultrasonic with vibration generated from the vibration board.

16. The pass box according to claim 15, comprising a control device configured to change the frequency and output of the ultrasound generated from the vibration board and/or to transmit the ultrasound intermittently to control a position or a moving speed of the mist for decontamination to concentrate said mist on the external surfaces of the article.
